Unit 2 Group Problem Session

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1 Instructions

Solve these problems together, in a group of four.
Each team member will write up the solution to one of the questions.
Submit your solutions to Gradescope (more instructions to follow in Canvas).
You will need to provide:

- The names of all team members
- Your solutions to the problems, indicating who wrote up each solution.
- **You may consult generative AI tools and other sources, but do not use them to write your solutions.**
- Cite any external sources or tools used and you used them, or indicate that you did not use any. **This part is required.**

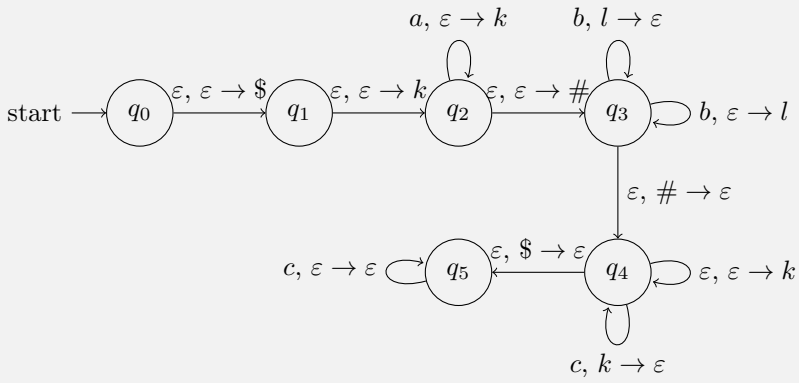
2 Questions

1. Let $\Sigma = \{a, b, c\}$. Draw a PDA that recognizes the following language:

$$\{a^k b^{2l} c^m \mid l \geq 0, m > k \geq 0\}$$

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PDA:



AI Usage and Outside Sources: No AI agent or tool was used during the solution of this problem, nor was the internet used in any way. The only thing referenced was the latex code for the Unit 1B group problem set for writing the the graph.

2. Write a CFG that generates the language

$$\{a^k b^{2l} c^m \mid l \geq k \geq 0, m \geq 0\}$$

Note this is NOT the language in the previous question.

The CFG does **not** have to be in Chomsky normal form (CNF).

3. Consider the following CFG.

$$\begin{aligned}
S &\to ZT \mid YR \mid ZZ \mid YY \mid 0 \mid 1 \\
X &\to ZT \mid YR \mid ZZ \mid YY \mid 0 \mid 1 \\
R &\to XY \\
T &\to XZ \\
Z &\to 0 \\
Y &\to 1
\end{aligned}$$

Consider the string 00100. Run the CYK algorithm to determine whether G generates this string. If it does, recover the parse tree that results from the algorithm. If not, say why not.

4. Consider the language $L = \{r\#st\#ts \mid r, s, t \in \{0,1\}^*\}$. Prove that L is not context free using the pumping lemma for context-free languages.